URBAN MOBILITY
RIDE BOOKING & DEMAND ANALYTICS

Data-Driven Insights & Interactive Excel Dashboard

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PROJECT OVERVIEW

The Big Picture

Turning massive raw data into actionable business intelligence.

THE GOAL

Analyze ride-booking trends in the NCR region and identify growth opportunities.

THE DATA

145,000+ booking records covering revenue, cancellations, and vehicle types.

THE OUTCOME

An interactive dashboard that reduces manual reporting and improves visibility.

THE PROBLEM STATEMENT

Operational Challenges

BUSINESS BOTTLENECKS

Messy Data

Raw data had duplicates, incorrect formats, and missing values, making it unreliable for decision-making.

Manual Effort

Reporting was slow and prone to human error, delaying critical business responses.

Hidden Losses

High cancellation rates were causing significant revenue loss without a clear 'Why' or actionable insights.

DATA CLEANING & PREPARATION

The Process

REMOVING NOISE

Deleted duplicate Booking IDs and handled missing values to ensure 100% data accuracy.

FORMATTING

Converted Date-Time columns and corrected numeric formats for revenue and distance metrics.

FEATURE ENGINEERING

Created 'Month' and 'Day' columns using the TEXT() function for granular trend analysis.

STRUCTURING

Organized raw data into an Excel Table for seamless Pivot Table and Dashboard integration.

TECHNICAL TOOLS USED

The Tech Stack

MICROSOFT EXCEL

The core platform for data cleaning, processing, and interactive dashboarding.

PIVOT TABLES & CHARTS

Used for summarizing 145k+ records and creating dynamic visual summaries.

XLOOKUP & FORMULAS

For efficient data retrieval and calculating derived metrics like Month and Day.

CONDITIONAL FORMATTING

To highlight critical issues like high wait times or low customer ratings instantly.

KEY BUSINESS METRICS

Project KPIs

₹51.4M

TOTAL REVENUE

Total booking value across all categories.

₹46.9M

SUCCESS REVENUE

Actual revenue realized from completed rides.

148.8K

TOTAL BOOKINGS

Massive scale of 145k+ records analyzed.

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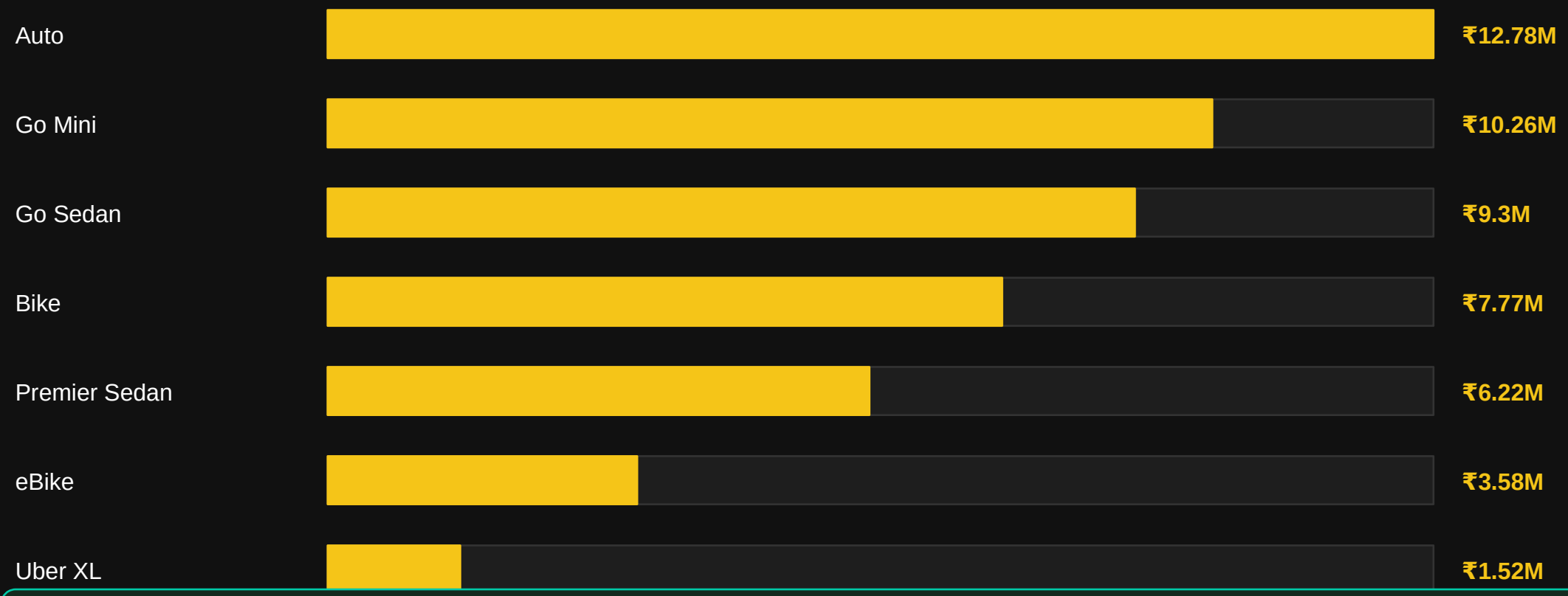
AVG. RATING

High customer satisfaction despite operational issues.

REVENUE & DEMAND BY VEHICLE

Market Analysis



AUTO DOMINANCE — Auto & Mini segments represent 45%+ of total booking value. Expanding the Auto fleet is the biggest growth opportunity.

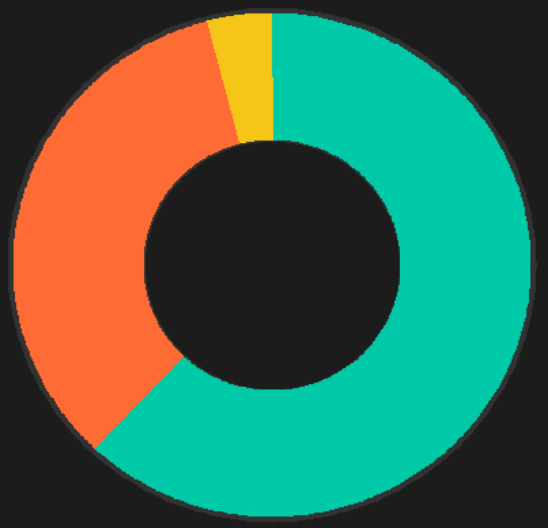
CANCELLATION ANALYSIS

Revenue Leakage

₹5.0M

ESTIMATED REVENUE LOSS

Booking Status Breakdown (K)



Completed Cancelled Incomplete

DRIVER BEHAVIOR

'Driver asked to cancel' and 'Driver not moving towards pickup' are the top operational issues.

LOCATION ISSUES

Wrong address or pickup location mismatches contribute to customer-side cancellations.

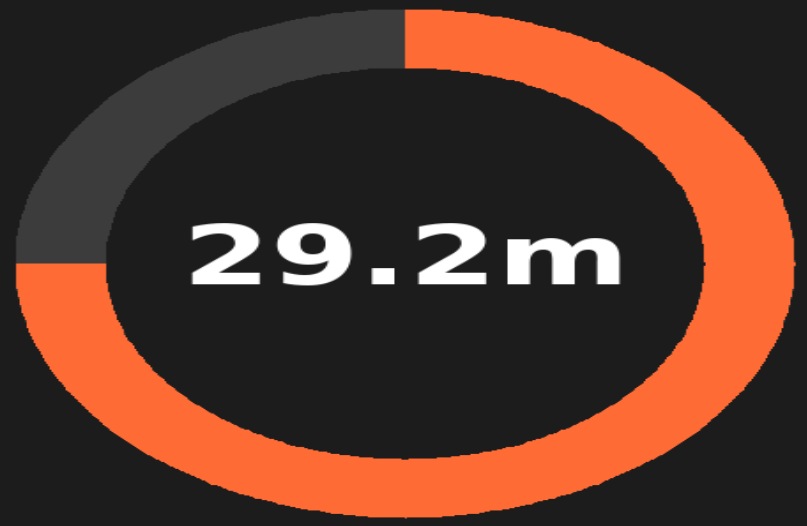
STRATEGIC ACTION

Implement driver penalties for forced cancellations and optimize dispatch logic to reduce wait times.

OPERATIONAL EFFICIENCY

Wait Time Analysis

AVG. CTAT



Customer Turnaround Time: High wait times lead to customer frustration and lower retention.

AVG. VTAT

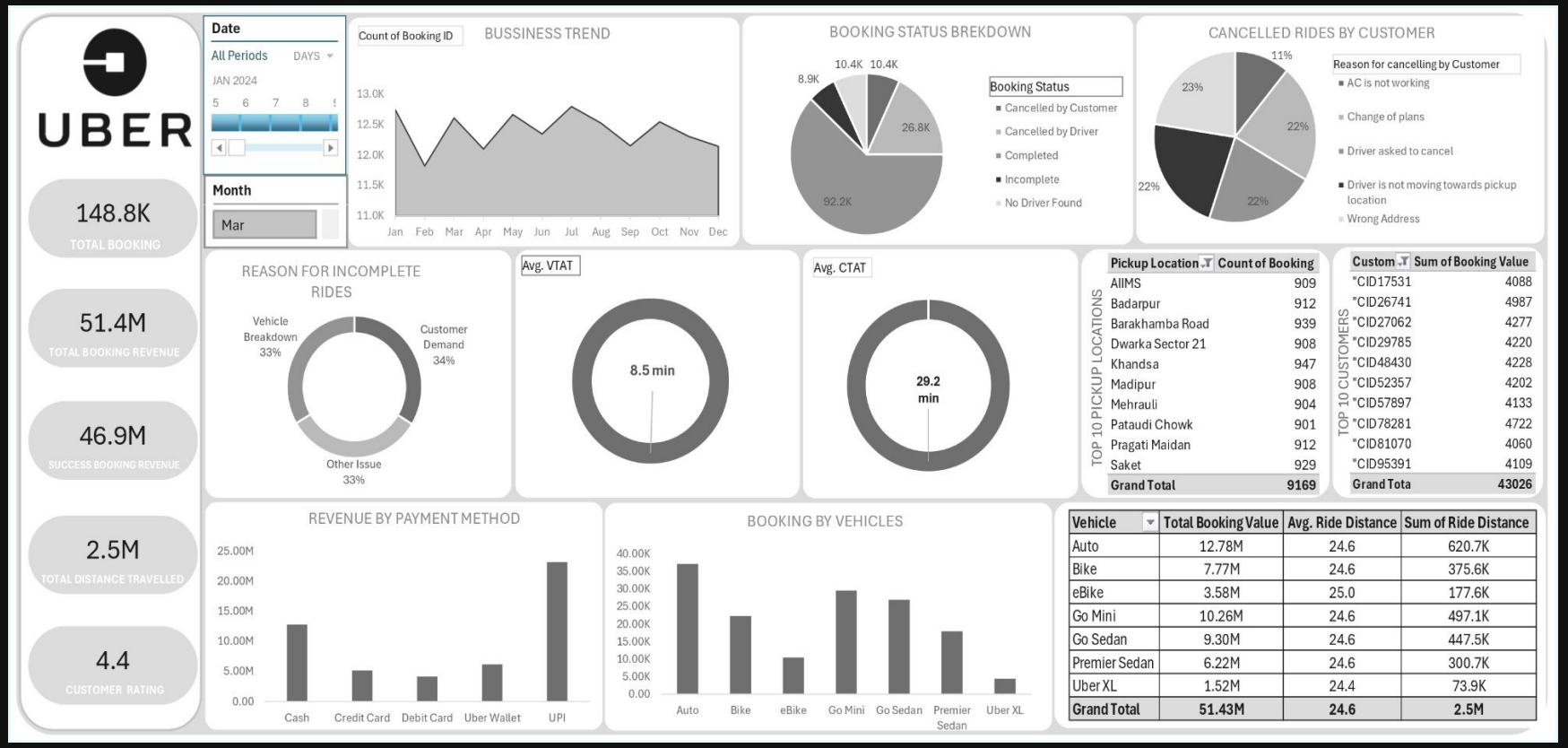
Vehicle Turnaround Time: Efficient driver response once a booking is accepted.

Target: Optimize dispatch logic to reduce CTAT below 20 minutes for better driver utilization.

INTERACTIVE DASHBOARD PREVIEW

Visual Summary

A one-stop view of Business Trends, Booking Status, and Revenue Metrics.



DYNAMIC SLICERS

Interactive Date & Month filters — drill into specific periods instantly.

MULTI-CHART VIEW

6+ charts for holistic operational & financial performance.

TOP PICKUP HOTSPOTS

AIIMS, Badarpur & Dwarka identified as highest demand locations.

FINAL RECOMMENDATIONS

Strategic Roadmap

REDUCE CANCELLATIONS

Implement penalties or incentives to stop 'Driver asked to cancel' issues and prevent heavy revenue loss.

FLEET EXPANSION

Increase the number of Autos to meet high demand and capture more market share in the NCR region.

OPTIMIZE CTAT

Improve routing and dispatch logic to reduce customer wait times and enhance the overall user experience.

MONITOR HOTSPOTS

Position more drivers near top pickup locations like AIIMS and Dwarka to align supply with customer demand.

CONCLUSION & LEARNINGS

Final Thoughts

"Data is the engine of modern business growth."

This project demonstrates how turning 145,000+ rows into a single interactive dashboard can make decision-making 10x faster and more accurate.

DATA-DRIVEN SUPPLY

Real-time monitoring helps align vehicle supply with demand hotspots like AIIMS and Dwarka.

THE NEXT STEPS

Automate the data pipeline for daily real-time updates and integrate predictive demand forecasting.